# MAT1100 Homework 3

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## Question 1

(a) *Proof.* Suppose that $G$ is a group of order 12 such that $n_3(G) = 4$. By Sylow's theorem $G$ has a Sylow 2-subgroup, to show that it is normal it suffices to show that $n_4(G) = 1$. Now $n_4(G) \mid 3$ and $n_4(G) = 1 \mod 2$.

Suppose for contradiction that $G$ has no normal Sylow 4-subgroup, then $n_4(G) = 3$. Since $n_3(G) = 4$ and each Sylow-3-subgroup of $G$ is cyclic, they contain 8 distinct order 3 elements. Now since $n_4(G) = 3$ and each subgroup is distinct, they contain at least 4 distinct even order elements. However, this would imply that $G$ contains at least 13 elements which is a contradiction.

Thus $G$ has a normal Sylow 2-subgroup $N$. There are only two groups of order 4, up to isomorphism, the Klein 4-group and the cyclic group of order 4.

I claim that $N$ is not cyclic. Suppose it was, then $G$ would be a semidirect product of $N = \langle x \rangle$ and $H = \langle y \rangle$ where $x$ has order 4 and $y$ has order 3.

Now $\mathrm{Aut}(N) \cong \mathbb{Z}_2$ as the only generators of $N$ are $x, x^3$ and an automorphism of a cyclic group is uniquely determined by the image of the generator.

However if $G = N \rtimes_\varphi H$ and $\varphi : H \to \mathrm{Aut}(N) \cong \mathbb{Z}_2$ then $\varphi$ must be trivial as the orders of the domain and codomain are relatively prime.

Thus, $G$ is a direct product of abelian groups and hence abelian which implies $H$ is normal. This contradicts the fact that $n_3(G) = 4$. Thus $N$ is isomorphic to the Klein 4-group.

Now suppose we fix an homomorphism $\varphi : H \to \mathrm{Aut}(N)$, the three non-identity elements of $N$ are indistinguishable so they can be permuted freely by an automorphism. Thus $\mathrm{Aut}(N) \cong S_3$. Now enumerate $N$ as $e, a, b, c$. $\varphi$ is determined by its image on $x$ the generator of $H$. Now $\varphi$ is non trivial and since $|x| = 3$ this implies that $|\varphi(x)| = 3$. I.e. it is a 3-cycle. Without loss of generality (up to relabeling of $N$) $\varphi_x(a) = b$, $\varphi_x(b) = c$ and $\varphi_x(c) = a$.

Now we define a homomorphism $f : G \to S_3$ specified by

$$x \mapsto (123)$$

and

$$N \mapsto \{e, (12)(34), (14)(23), (13)(24)\}$$

and it is clear that

$$\begin{aligned}(123)(12)(34)(123)^{-1} &= (14)(23)\\ (123)(14)(23)(123)^{-1} &= (13)(24)\\ (123)(13)(24)(123)^{-1} &= (12)(34)\end{aligned}$$

thus

$$f\varphi_x(k) = (123)f(k)(123)^{-1}$$

for any $k \in N$ and therefore for any $k, \ell \in N$:

$$\begin{aligned}f(kx^i\ell x^j) &= f(k\varphi_{x^i}(\ell)x^{i+j})\\ &= f(k)(123)^i f(\ell)(123)^{-i}(123)^{i+j}\\ &= f(k)(123)^i f(\ell)(123)^j\\ &= f(kx^i)f(\ell x^j)\end{aligned}$$

hence $f$ is a homomorphism. Additionally, it is clearly injective. Thus $G \cong A_4$. □

(b) *Proof.* By the last question $G \cong A_4$ and $G = N \rtimes_\varphi H$ such that every element of $N$ can be written uniquely as $kx^i$ for $k \in N$. Now under the map $f$ as defined above this implies that every element of $A_4$ can be written as uniquely as $vc^i$ for $v \in V$ and $c = (123)$ with $i \in \{0, 1, 2\}$. Now given $wc^j \in A_4$ we can write

$$vc^iwc^j = vc^iwc^{-i}c^ic^j$$

and since $w = w^{-1}$

$$c^2wc^{-2} = c^-1wc = (cwc^{-1})^{-1} = cwc^{-1}$$

then with the rule

$$\begin{aligned}(123)(12)(34)(123)^{-1} &= (14)(23)\\ (123)(14)(23)(123)^{-1} &= (13)(24)\\ (123)(13)(24)(123)^{-1} &= (12)(34)\end{aligned}$$

the multiplication is fully determined.

Now $A_4$ is not the direct product of $V$ and $C_3$ as both of these groups are abelian and this would imply $A_4$ is too. However, $A_4$ is not abelian as

$$(123)(12)(34)(123)^{-1} = (14)(23) \neq (12)(34)$$

□